

Zero-Shot Linear Quadratic Regulation from Reward-Free Data: Structured Successor Features with Stability and Finite-Sample Guarantees

Quang Huy Dao^a, Khanh Thai^a and Phuong Nam Dao^{a,*}

^a*School of Electrical and Electronic Engineering, Hanoi University of Science and Technology, Hanoi, Vietnam*

ARTICLE INFO

Keywords:

Zero-shot reinforcement learning
Linear quadratic regulator
Successor features
Adaptive dynamic programming
Data-driven control
Persistency of excitation

ABSTRACT

Reinforcement learning (RL) and adaptive dynamic programming (ADP) algorithms for the linear quadratic regulator (LQR) learn the optimal gain for *one* performance index: whenever the designer changes the weights (Q, R) , data must be re-collected or re-processed and the policy iteration re-run. Zero-shot RL, in contrast, learns from reward-free interaction a representation from which the optimal policy for *any* reward revealed at test time can be read out without further learning, but existing zero-shot methods rely on neural approximations and offer no stability or optimality guarantee. This paper brings zero-shot RL to linear optimal control. We show that for linear systems with quadratic costs the successor features of every linear policy are exact quadratic forms, that the policy-evaluation map $(Q, R) \mapsto Q$ -function kernel is linear and given by a closed-form successor operator, and that a fixed-size second-moment statistic of reward-free data is a sufficient zero-shot representation. On this basis we propose a zero-shot LQR algorithm for discrete-time systems: an offline phase compresses arbitrary reward-free trajectories into the representation, and a test-time phase maps any task (Q, R) , given explicitly or through reward labels, to a gain through a data-free policy iteration, or with no iteration at all through successor-feature generalized policy improvement (GPI). We prove that with noise-free persistently exciting data the returned gains are exactly the Riccati-optimal ones for all tasks simultaneously; that the GPI policy is exponentially stabilizing and no worse than any library policy; and that under sub-Gaussian process noise, with probability $1 - \delta$ and uniformly over a compact task set, the gains are stabilizing and their relative suboptimality is $O(\sigma_w \sqrt{\log(1/\delta)/(N \lambda_{PE})})$, where N is the number of samples and λ_{PE} the persistency-of-excitation level. A data-computable certificate verifies stability and suboptimality for each task. The framework extends to continuous-time systems through an integral-data Gram representation and a data-free Kleinman iteration. Simulations on the unstable batch reactor confirm machine-precision exactness over 200 random tasks, test-time cost independent of the data length (0.4 ms versus 14–359 ms for per-task off-policy Q-learning), and the predicted sample complexity.

1. Introduction

Reinforcement learning (RL) and adaptive dynamic programming (ADP) have made it possible to solve linear quadratic regulation (LQR) problems without knowledge of the system matrices [27, 21, 19, 28]. For discrete-time (DT) systems $x_{k+1} = Ax_k + Bu_k$, on- and off-policy Q-learning [7, 6, 25, 20, 32, 44] implement Hwer's policy iteration [16] from measured state–input data; for continuous-time (CT) systems $\dot{x} = Ax + Bu$, integral RL and off-policy ADP [40, 18, 39, 31] implement Kleinman's algorithm [22]. A parallel line of work on direct data-driven control [11, 41, 13, 8], built on Willems' fundamental lemma [43], and learning-theoretic analyses of LQR [12, 29, 38, 24, 14, 2] characterize how much data such methods need.

All of these methods solve the problem for *one fixed* performance index. The weights (Q, R) appear in the Bellman equations that are solved from data, so when the designer changes them—to trade tracking accuracy against actuator effort, to adapt to a new mission phase, or simply during controller tuning—the policy iteration has to be run again on the stored data, or new data have to be collected when the method is on-policy. In practice the weights are rarely known in advance: they are tuned iteratively, and every tuning step then costs a full learning run.

*Corresponding author

✉ nam.daophuong@hust.edu.vn (P.N. Dao)
ORCID(s): 0000-0002-8333-5572 (P.N. Dao)

The machine-learning community has studied the analogous question under the name *zero-shot RL* [36, 37]: an agent interacts with the environment *without any reward*, builds a representation, and at test time must output a near-optimal policy for *any* reward function, instantly and without further learning or planning in the environment. The two main families are successor features (SF) [10, 3, 5], which assume rewards linear in known features and combine the SFs of several policies through generalized policy improvement (GPI), and forward–backward (FB) representations [36, 4, 37], which learn a low-rank factorization $M^{\pi_z}(s, a, s') \approx F(s, a, z)^\top B(s')$ of the successor measure together with a task-conditioned policy family π_z . Follow-up work improves representation learning [30] and robustness to low-quality data [17]. These methods are evaluated on simulated locomotion and navigation benchmarks, rely on neural network approximation, and provide no closed-loop stability guarantee or error bound in terms of the amount and quality of data—the two properties that make a controller acceptable in control engineering.

Closer to control theory, multitask and meta-learning formulations of LQR have also been studied. Multitask LQG control [34] learns a single shared stabilizing controller across a distribution of systems and objectives, with generalization bounds that grow with a bisimulation-based heterogeneity measure between tasks; the controller is common to all tasks and is therefore biased away from each task-specific optimum. Meta-learning approaches such as MAML-LQR [15, 35] instead learn an initialization from which a task-specific gain is reached after further policy-gradient adaptation on the new task, so a new task still requires online learning. A task-space-complexity framework for zero-shot generalization [9] uses LQR as one of its running examples but measures transfer through a performance-divergence metric between source and target contexts rather than through a closed-form, exact policy map. None of these methods returns, from a single reward-free dataset and without further interaction or adaptation, the exact gain for every task in a continuum, or a stability certificate for it; this is what we do next.

To the authors' knowledge, zero-shot RL in the representation-based sense above—a reward-free representation from which a near-optimal policy for *any* task is read out through a closed-form map, with provable stability and suboptimality guarantees—has not been formulated or analyzed for the LQR problem; the closest control-theoretic uses of the term “zero-shot” rely instead on a performance-divergence measure between a fixed source policy and unseen targets [9], with no representation and no closed form. Conversely, none of the ADP or data-driven LQR methods above is zero-shot in the sense of amortizing learning across all performance indices. This paper connects the two lines. Our starting observation is that the linear–quadratic structure makes the objects that zero-shot RL approximates *exactly* finite-dimensional: with the quadratic features $\phi(z) = z \otimes z$, $z = [x^\top, u^\top]^\top$, every stage cost $x^\top Q x + u^\top R u$ is linear in ϕ , the successor features of any linear policy are quadratic forms, and the map from the task to the policy's Q-function kernel is a linear operator in closed form. The contributions are as follows.

1. **Problem and exact representation.** We formulate the zero-shot LQR problem (Problem 1) and show (Proposition 1) that the successor features of LQR admit an exact forward–backward factorization, with a closed-form successor operator $\mathcal{L}_K$ that is linear in the task. A fixed-size second-moment statistic of reward-free data, whose size depends neither on the number of samples nor on the task, suffices to evaluate every policy on every task.
2. **Algorithms.** For DT systems we propose a two-phase algorithm: an offline phase that compresses arbitrary reward-free, off-policy trajectories (Algorithm 1), and a test-time phase that maps any task, given as weights or as reward labels, to a gain by a data-free policy iteration or, with no iteration, by SF-GPI over a policy library (Algorithm 2). For CT systems an integral-data Gram representation supports a data-free Kleinman iteration (Algorithm 3).
3. **Guarantees.** With noise-free persistently exciting (PE) data the returned gains equal the Riccati solutions for all tasks simultaneously, every iterate is stabilizing, and the iterates coincide with per-task off-policy Q-learning, which, unlike our method, needs quadratic rather than linear PE (Theorem 1). The SF-GPI policy is exponentially stabilizing and never worse than the best library policy (Theorem 2). Under sub-Gaussian process noise, with probability $1 - \delta$, the gains are stabilizing and the relative suboptimality is bounded uniformly over a compact task set by a quantity of order $\sigma_w \sqrt{\log(1/\delta)/(N \lambda_{PE})}$; a data-computable certificate verifies both properties for each individual task (Theorem 3). The CT counterpart is Theorem 4.
4. **Evidence.** On the open-loop unstable batch reactor we verify exactness over 200 random tasks, a test-time cost independent of the data length, the predicted $1/N$ decay of suboptimality for three PE levels, the near-optimality of zero-iteration SF-GPI, the CT extension, and task inference from noisy reward labels.

We state plainly what the structure implies. For LQR the zero-shot representation turns out to be equivalent to the least-squares estimate of the one-step transition operator $\Theta = [A \ B]$, so the test-time phase can be read as certainty-equivalent control computed on demand. We regard this as a finding rather than a weakness: it shows that the objects

FB and SF methods learn with neural networks collapse, in the linear–quadratic case, to closed forms whose error can be controlled, and it gives the first finite-sample, closed-loop guarantees for zero-shot RL on continuous state and action spaces. The contributions that go beyond certainty equivalence are the operator viewpoint (Proposition 1), the zero-iteration SF-GPI controller with its stability proof, the uniform-over-tasks guarantee obtained from a *single* probabilistic event (no union bound over tasks), and the per-task data-computable certificate.

Organization. Section 2 states the problem. Section 3 derives the structured successor features. Section 4 presents the DT algorithm and its analysis, and Section 5 the CT extension. Section 6 reports simulations and Section 7 discusses limitations. Proofs are collected in the Appendix.

Notation. $A > 0$ (≥ 0) denotes a positive (semi)definite matrix, $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ the extreme eigenvalues of a symmetric matrix, $\|\cdot\|$ the Euclidean norm or the induced spectral norm, $\rho(\cdot)$ the spectral radius, and $\otimes$ the Kronecker product. $\text{vec}(\cdot)$ stacks columns, so that $\text{vec}(XYZ) = (Z^\top \otimes X) \text{vec}(Y)$ and $z^\top W z = \text{vec}(W)^\top (z \otimes z)$. A matrix M is Schur if $\rho(M) < 1$ and Hurwitz if all its eigenvalues have negative real part. $\text{blkdiag}(\cdot)$ is a block-diagonal matrix. I_n is the $n \times n$ identity.

2. Problem formulation

2.1. System, tasks and costs

Consider the DT linear system

$$x_{k+1} = Ax_k + Bu_k + w_k = \Theta z_k + w_k, \quad z_k := \begin{bmatrix} x_k \\ u_k \end{bmatrix} \in \mathbb{R}^d, \quad (1)$$

with state $x_k \in \mathbb{R}^n$, input $u_k \in \mathbb{R}^m$, $d := n + m$, *unknown* $\Theta := [A \ B] \in \mathbb{R}^{n \times d}$, and process noise w_k (possibly zero). A *task* is a weight pair (Q, R) collected in

$$W := \text{blkdiag}(Q, R) \in \mathbb{R}^{d \times d}, \quad c(W) := \lambda_{\min}(W) = \min\{\lambda_{\min}(Q), \lambda_{\min}(R)\}. \quad (2)$$

For a linear policy $u = -Kx$ let $\Pi_K := [I_n \ -K^\top]^\top \in \mathbb{R}^{d \times n}$, so that the closed-loop matrix is $A - BK = \Theta \Pi_K$. The set of stabilizing gains is $\mathcal{K}(\Theta) := \{K : \rho(\Theta \Pi_K) < 1\}$. For $K \in \mathcal{K}(\Theta)$ the value matrix $P_K(W)$ is the unique solution of the Lyapunov equation

$$P_K = (A - BK)^\top P_K (A - BK) + Q + K^\top R K = (\Theta \Pi_K)^\top P_K (\Theta \Pi_K) + \Pi_K^\top W \Pi_K, \quad (3)$$

and $x_0^\top P_K x_0 = \sum_{t \geq 0} (x_t^\top Q x_t + u_t^\top R u_t)$ along the noise-free closed loop. We measure performance by

$$J_W(K) := \text{tr } P_K(W), \quad (4)$$

the expected cost from $x_0 \sim \mathcal{N}(0, I_n)$, which also equals the average cost under unit-covariance process noise. The optimal value $P^*(W)$ is the stabilizing solution of the discrete algebraic Riccati equation (DARE)

$$P = A^\top P A - A^\top P B (R + B^\top P B)^{-1} B^\top P A + Q, \quad (5)$$

with optimal gain $K^*(W) = (R + B^\top P^* B)^{-1} B^\top P^* A$ and $J_W^* = \text{tr } P^*(W)$. The associated Q-function kernel of a policy K is

$$H_K(W) := W + \Theta^\top P_K(W) \Theta = \begin{bmatrix} H^{xx} & H^{xu} \\ H^{ux} & H^{uu} \end{bmatrix}, \quad Q_K^W(x, u) = z^\top H_K(W) z. \quad (6)$$

2.2. Zero-shot LQR

Assumption 1 (Tasks). (A, B) is stabilizable. The task set $\mathcal{W}$ is a compact set of weights $W = \text{blkdiag}(Q, R)$ with $Q > 0$, $R > 0$, and $c_{\mathcal{W}} := \min_{W \in \mathcal{W}} c(W) > 0$.

Assumption 2 (Reward-free data). A dataset $\mathcal{D} = \{(x_k, u_k, x_{k+1})\}_{k=0}^{N-1}$ is generated by (1), possibly as several concatenated episodes, with inputs chosen by an arbitrary causal exploration rule (open-loop excitation, any behavior policy, or a mixture). No cost or reward is observed. Let $\mathcal{F}_k$ be the σ -algebra generated by $z_0, \dots, z_k, w_0, \dots, w_{k-1}$; then w_k is conditionally zero-mean given $\mathcal{F}_k$, and each coordinate of w_k is conditionally σ_{w_k} -sub-Gaussian.

Define the data matrices $Z := [z_0, \dots, z_{N-1}] \in \mathbb{R}^{d \times N}$ and $X^+ := [x_1, \dots, x_N] \in \mathbb{R}^{n \times N}$ and the empirical second moments

$$\hat{\Sigma}_{zz} := \frac{1}{N} Z Z^\top, \quad \hat{\Sigma}_{x+z} := \frac{1}{N} X^+ Z^\top. \quad (7)$$

Definition 1 (PE level). *The data are persistently exciting (PE) if $\text{rank } Z = d$, equivalently $\hat{\Sigma}_{zz} > 0$. The PE level is $\lambda_{\text{PE}} := \lambda_{\min}(\hat{\Sigma}_{zz})$.*

Problem 1 (Zero-shot LQR). *Offline phase: from $\mathcal{D}$ alone, compute a representation $\mathcal{R} = \mathcal{R}(\mathcal{D})$ whose size is independent of N and of any task. Test phase: for any $W \in \mathcal{W}$, revealed either explicitly or through reward labels $r_k = z_k^\top W z_k$ (possibly noisy) on $\mathcal{D}$, return a gain $\hat{K}(W) = \Gamma(\mathcal{R}, W)$ without interacting with the system and without reprocessing $\mathcal{D}$ (apart from one linear pass to infer W when only labels are given), such that*

- (i) $\hat{K}(W) \in \mathcal{K}(\Theta)$ (closed-loop stability);
- (ii) $J_W(\hat{K}(W)) \leq (1 + \eta_N) J_W^*$ with $\sup_{W \in \mathcal{W}} \eta_N \rightarrow 0$ as $N \rightarrow \infty$ (uniform near-optimality);
- (iii) the computational cost of Γ is independent of N .

Classical off-policy Q-learning [7, 20] satisfies (i)–(ii) for one task but violates (iii): every task requires a new policy iteration in which each policy-evaluation step solves a least-squares problem over all N samples. On-policy methods violate the offline requirement as well, since each task requires new interaction.

3. Structured successor features for LQR

Zero-shot RL rests on the successor representation [10]. For a policy π , the successor features with respect to a feature map ϕ are $\psi^\pi(z) = \sum_{t \geq 0} \phi(z_t)$ along the trajectory started at $z_0 = z$ [3]. If the reward is linear in the features, $r = \phi^\top w$, then $Q_r^\pi = \psi^\pi^\top w$ for every w , so evaluating a known policy on a new task reduces to an inner product. FB representations [36] go one step further and learn a family of policies π_ζ together with a factorization $Q_{r^\zeta}^\pi(z) = F(z, \zeta)^\top \zeta$, where the task code $\zeta = \mathbb{E}[B(z)r(z)]$ is computed by a *backward* map; at test time π_ζ is read out directly.

For LQR we use the quadratic features and task code

$$\phi(z) := z \otimes z \in \mathbb{R}^{d^2}, \quad w := \text{vec}(W), \quad z^\top W z = \phi(z)^\top w. \quad (8)$$

Define the state–action transition matrix of the noise-free closed loop under K ,

$$G_K := \Pi_K \Theta \in \mathbb{R}^{d \times d}, \quad z_{t+1} = G_K z_t \text{ when } u_t = -K x_t, \quad t \geq 1. \quad (9)$$

Lemma 1. *For every K , the nonzero eigenvalues of G_K coincide with those of $A - BK$, so $\rho(G_K) = \rho(A - BK)$. For $K \in \mathcal{K}(\Theta)$, the kernel (6) is the unique solution of*

$$H_K(W) = W + G_K^\top H_K(W) G_K, \quad (10)$$

and $P_K(W) = \Pi_K^\top H_K(W) \Pi_K$.

Proposition 1 (Exact successor features and successor operator). *Let $K \in \mathcal{K}(\Theta)$ and consider the noise-free closed loop. Then:*

- (a) the successor features of K are $\psi^K(z) = (I_{d^2} - G_K \otimes G_K)^{-1}(z \otimes z)$, a quadratic form in z ;
- (b) for every task, $Q_K^W(z) = \psi^K(z)^\top \text{vec}(W)$ and $\text{vec}(H_K(W)) = \mathcal{L}_K \text{vec}(W)$ with the successor operator

$$\mathcal{L}_K := (I_{d^2} - G_K^\top \otimes G_K^\top)^{-1}; \quad (11)$$

- (c) G_K , and hence $\mathcal{L}_K$ and ψ^K , depend on the data-generating system only through Θ .

Proposition 1 says that in LQR the Q-function of any linear policy is *linear in the task*, with an explicit operator. It maps onto the FB template: the backward map is the least-squares projection of rewards on ϕ ,

$$\hat{w} = \left(\sum_k \phi(z_k) \phi(z_k)^\top \right)^\dagger \sum_k \phi(z_k) r_k, \quad (12)$$

which recovers w exactly from noise-free labels whenever the quadratic features of the data span the symmetric matrices; the forward map is $F(z, W) = \psi^{K^*(W)}(z)$, so that $Q_W^*(z) = F(z, W)^\top \text{vec}(W)$. Unlike generic FB, neither map has to be approximated: ψ^K is available in closed form once Θ is, and $K^*(W)$ is reached by policy iteration in which every evaluation is (10). Proposition 1(c) identifies what the offline phase must learn: the one-step transition operator Θ . When the data are PE,

$$\hat{\Theta} := \hat{\Sigma}_{x+z} \hat{\Sigma}_{zz}^{-1} = X^+ Z^\top (Z Z^\top)^{-1} \quad (13)$$

is its least-squares estimate. The pair $\mathcal{R} := (\hat{\Sigma}_{zz}, \hat{\Sigma}_{x+z})$ has $d^2 + nd$ entries regardless of N and is therefore an admissible zero-shot representation. We write $\hat{G}_K := \Pi_K \hat{\Theta}, \hat{\mathcal{L}}_K, \hat{H}_K(W)$ and $\hat{P}_K(W)$ for the quantities computed with $\hat{\Theta}$ in place of Θ .

Remark 1 (Why linear rather than quadratic regression). *Off-policy Q-learning [7, 20] evaluates K by regressing $z_k^\top W z_k$ on $\phi(z_k) - \phi(\Pi_K x_{k+1})$, which needs the quadratic PE condition $\text{rank}[\phi(z_0), \dots, \phi(z_{N-1})] = d(d+1)/2$, and therefore at least $d(d+1)/2$ samples. Estimating the linear operator Θ needs only $\text{rank } Z = d$. Under process noise the quadratic regression is moreover biased by the term $\text{tr}(H \text{Cov}(w))$, whereas (13) is consistent.*

4. Zero-shot LQR for discrete-time systems

4.1. Algorithms

The offline phase (Algorithm 1) accumulates the second moments. It optionally precomputes a *policy library*: M gains $K_1, \dots, K_M$ that stabilize $\hat{\Theta}$ (for instance $\hat{K}(W_i)$ for anchor tasks $W_1, \dots, W_M$) together with their successor operators $\hat{\mathcal{L}}_{K_i}$. The test phase (Algorithm 2) supports two modes.

Mode PI (policy iteration in representation space). Starting from a gain $K_0 \in \mathcal{K}(\hat{\Theta})$, iterate

$$\hat{H}_j = W + \hat{G}_{K_j}^\top \hat{H}_j \hat{G}_{K_j}, \quad K_{j+1} = (\hat{H}_j^{uu})^{-1} \hat{H}_j^{ux}. \quad (14)$$

Each step solves a $d \times d$ Lyapunov equation and touches no data. The initial gain can be taken from the library as $K_0 = \arg \min_{K_i} \text{tr}(\Pi_{K_i}^\top \hat{H}_{K_i}(W) \Pi_{K_i})$, which costs M matrix–vector products with the precomputed $\hat{\mathcal{L}}_{K_i}$. This removes the usual requirement of a task-specific initial stabilizing gain.

Mode GPI (no iteration). For each library policy compute $\hat{H}_i = \hat{H}_{K_i}(W)$ from $\text{vec}(\hat{H}_i) = \hat{\mathcal{L}}_{K_i} \text{vec}(W)$, the improved gain $K_i^+ = (\hat{H}_i^{uu})^{-1} \hat{H}_i^{ux}$, and the value kernel $S_i = \hat{H}_i^{xx} - \hat{H}_i^{xu} K_i^+$. Apply the switched linear policy

$$\pi_W^{\text{GPI}}(x) = -K_{i^*(x)}^+ x, \quad i^*(x) \in \arg \min_i x^\top S_i x. \quad (15)$$

This is exactly GPI [3], $\pi(x) \in \arg \min_u \min_i Q_{K_i}^W(x, u)$, written in closed form. Its test-time cost is M products of a $d^2 \times d^2$ matrix with a vector.

4.2. Exact zero-shot optimality with noise-free data

Theorem 1 (Exactness). *Let $w_k \equiv 0$, let the data be PE, and let (A, B) be stabilizable. Then $\hat{\Theta} = \Theta$, and for every W with $Q > 0$, $R > 0$ and every $K_0 \in \mathcal{K}(\Theta)$ the iteration (14) satisfies:*

- (a) $K_j \in \mathcal{K}(\Theta)$ for all j ;
- (b) $P^*(W) \leq P_{K_{j+1}}(W) \leq P_{K_j}(W)$;
- (c) $K_j \rightarrow K^*(W)$ and $P_{K_j}(W) \rightarrow P^*(W)$ quadratically.

If in addition the quadratic features of the data span the symmetric matrices, the iterates coincide with those of per-task off-policy Q-learning [7, 20] run on $\mathcal{D}$ with the same W and K_0 , and the backward map (12) recovers W exactly from noise-free labels.

Algorithm 1 Offline phase: reward-free representation (DT)**Require:** reward-free data $\{(x_k, u_k, x_{k+1})\}_{k=0}^{N-1}$; optional anchor tasks $W_1, \dots, W_M$

- 1: $\hat{\Sigma}_{zz} \leftarrow \frac{1}{N} \sum_k z_k z_k^\top$, $\hat{\Sigma}_{x+z} \leftarrow \frac{1}{N} \sum_k x_{k+1} z_k^\top$ $\triangleright$ streaming, $O(d^2)$ memory
- 2: check PE: $\lambda_{\text{PE}} = \lambda_{\min}(\hat{\Sigma}_{zz}) > 0$; $\hat{\Theta} \leftarrow \hat{\Sigma}_{x+z} \hat{\Sigma}_{zz}^{-1}$
- 3: **for** $i = 1, \dots, M$ **do** $\triangleright$ optional policy library
- 4: $K_i \leftarrow$ Mode PI of Algorithm 2 (or DARE of $\hat{\Theta}$) for W_i
- 5: $\hat{L}_{K_i} \leftarrow (I - \hat{G}_{K_i}^\top \otimes \hat{G}_{K_i}^\top)^{-1}$
- 6: **end for**
- 7: **return** $\mathcal{R} = (\hat{\Sigma}_{zz}, \hat{\Sigma}_{x+z})$ and library $\{(K_i, \hat{L}_{K_i})\}_{i=1}^M$; the raw data may be discarded

Algorithm 2 Test phase: zero-shot gain for a new task (DT)**Require:** representation $\mathcal{R}$, library, task W or labels $\{r_k\}$; accuracy ϵ_{tol} ; optional noise level $\bar{\epsilon}$

- 1: **if** only labels are given **then**
- 2: $W \leftarrow$ backward map (12); project Q, R onto $\{\geq \epsilon I\}$
- 3: **end if**
- 4: **for** $i = 1, \dots, M$ **do**
- 5: $\text{vec}(\hat{H}_i) \leftarrow \hat{L}_{K_i} \text{vec}(W)$
- 6: **end for**
- 7: **if** Mode GPI **then**
- 8: **return** switched policy (15)
- 9: **end if**
- 10: $K_0 \leftarrow \arg \min_i \text{tr}(\Pi_{K_i}^\top \hat{H}_i \Pi_{K_i})$, $j \leftarrow 0$
- 11: **repeat**
- 12: $\hat{H}_j \leftarrow$ solve $\hat{H}_j = W + \hat{G}_{K_j}^\top \hat{H}_j \hat{G}_{K_j}$; $K_{j+1} \leftarrow (\hat{H}_j^{uu})^{-1} \hat{H}_j^{ux}$; $j \leftarrow j + 1$
- 13: **until** $\|K_j - K_{j-1}\| < \epsilon_{\text{tol}}$
- 14: $\hat{K} \leftarrow K_j$, $\hat{P} \leftarrow \Pi_{\hat{K}}^\top \hat{H}_{\hat{K}} \Pi_{\hat{K}}$
- 15: **certificate:** if $\bar{\epsilon} \leq c(W)/(12\|\hat{P}\|)$, report “stabilizing, $J_W(\hat{K}) \leq (1 + 20\bar{\epsilon}\|\hat{P}\|/c(W))J_W^*$ ” $\triangleright$ Theorem 3
- 16: **return** $\hat{K}$

The theorem holds for all tasks at once: a single reward-free dataset, compressed once, solves every LQR problem on the plant exactly. Only linear PE is needed, which requires $N \geq d$ samples, compared with $N \geq d(d+1)/2$ for per-task Q-learning.

Theorem 2 (Zero-iteration SF-GPI). *Let $\hat{\Theta} = \Theta$ (e.g., under the hypotheses of Theorem 1), let $K_1, \dots, K_M \in \mathcal{K}(\Theta)$, and let W satisfy $Q > 0, R > 0$. Define $V(x) := \min_i x^\top S_i x$. Then along the closed loop $x_{t+1} = Ax_t + B\pi_W^{\text{GPI}}(x_t)$:*

- (a) $V(x_{t+1}) \leq V(x_t) - x_t^\top Q x_t$, and $\lambda_{\min}(Q)\|x\|^2 \leq V(x) \leq \max_i \|S_i\| \|x\|^2$; hence the origin is globally exponentially stable, with $V(x_t) \leq (1 - \lambda_{\min}(Q)/\max_i \|S_i\|)^t V(x_0)$;
- (b) the accumulated cost satisfies $\sum_{t \geq 0} (x_t^\top Q x_t + u_t^\top R u_t) \leq V(x_0) \leq \min_i x_0^\top P_{K_i}(W) x_0$ for every x_0 .

For $M = 1$, SF-GPI reduces to one Hwerner improvement step from K_1 . For $M > 1$ it is no worse, from every initial state, than the best library policy for that state, without solving any equation at test time.

4.3. Finite-sample guarantees with noisy data

Let $\bar{z}^2 \geq \lambda_{\max}(\hat{\Sigma}_{zz})$ and $\lambda_0 \leq \lambda_{\text{PE}}$ be deterministic levels, and define

$$\bar{\epsilon}_N(\delta) := \sqrt{\frac{4n\sigma_w^2}{N\lambda_0} \left(\frac{d}{2} \log(1 + \bar{z}^2/\lambda_0) + \log \frac{n}{\delta} \right)}. \quad (16)$$

Lemma 2 (Representation error). *Under Assumption 2, for any $\delta \in (0, 1)$ the event*

$$\mathcal{E} := \{\lambda_0 I \leq \hat{\Sigma}_{zz} \leq \bar{z}^2 I\} \cap \{\|\hat{\Theta} - \Theta\| > \bar{\epsilon}_N(\delta)\}$$

has probability at most δ .

Lemma 3 (One-sided perturbation). *Let $\Theta_1, \Theta_2 \in \mathbb{R}^{n \times d}$ with $\|\Theta_1 - \Theta_2\| \leq \varepsilon$, let W satisfy $c = c(W) > 0$, let K stabilize Θ_1 , and let P_1 be its value matrix on Θ_1 with $p_1 = \|P_1\|$. If $\varepsilon \leq c/(6p_1)$, then K stabilizes Θ_2 , and its value matrix on Θ_2 satisfies $P_2 \leq (1 + 6\varepsilon p_1/c)P_1$.*

Theorem 3 (Uniform zero-shot guarantee and certificate). *Let Assumptions 1–2 hold, fix $\delta \in (0, 1)$, and let $\hat{K}(W)$, $\hat{P}(W)$ be the output of Algorithm 2 (Mode PI run to convergence) with $\bar{\varepsilon} = \bar{\varepsilon}_N(\delta)$. With probability at least $1 - \delta$, whenever $\lambda_0 I \preceq \hat{\Sigma}_{zz} \preceq \bar{z}^2 I$, the following hold simultaneously for all $W \in \mathcal{W}$.*

(a) Certificate. *If $\bar{\varepsilon}_N(\delta) \leq c(W)/(12\|\hat{P}(W)\|)$, then $\hat{K}(W) \in \mathcal{K}(\Theta)$ and*

$$\begin{aligned} P^\star(W) &\leq P_{\hat{K}}(W) \leq \left(1 + 20\bar{\varepsilon}_N \frac{\|\hat{P}\|}{c(W)}\right) P^\star(W), \\ \|\hat{P} - P^\star\| &\leq 14\bar{\varepsilon}_N \frac{\|\hat{P}\|^2}{c(W)}, \quad \|\hat{K} - K^\star\| \leq \frac{\|\hat{P}\|}{c(W)} \sqrt{30\bar{\varepsilon}_N}. \end{aligned} \quad (17)$$

(b) Uniform rate. *Let $\bar{p}_{\mathcal{W}} := \max_{W \in \mathcal{W}} \|P^\star(W)\|$. If $\bar{\varepsilon}_N(\delta) \leq c_{\mathcal{W}}/(18\bar{p}_{\mathcal{W}})$, then every $W \in \mathcal{W}$ satisfies the certificate in (a), and*

$$\sup_{W \in \mathcal{W}} \frac{J_W(\hat{K}(W)) - J_W^\star}{J_W^\star} \leq 27 \frac{\bar{p}_{\mathcal{W}}}{c_{\mathcal{W}}} \bar{\varepsilon}_N(\delta) = O\left(\frac{\bar{p}_{\mathcal{W}}}{c_{\mathcal{W}}} \sigma_w \sqrt{\frac{n(d \log(1 + \bar{z}^2/\lambda_0) + \log(n/\delta))}{N \lambda_0}}\right). \quad (18)$$

Remark 2 (Role of data and excitation). *The bound (18) decreases as $1/\sqrt{N\lambda_0}$: halving the error requires either four times more data or four times more excitation. Since λ_{PE} and $\bar{z}^2$ are computed from $\hat{\Sigma}_{zz}$, both the bound and the certificate in (a) can be evaluated from the representation alone, given σ_w . Only one probabilistic event is involved, so the guarantee holds for all tasks at once, with no union bound over $\mathcal{W}$, which is what makes it a zero-shot guarantee.*

Remark 3 (Tightness). *The rate in (18) is linear in $\bar{\varepsilon}_N$, which is enough for a clean, fully explicit and data-computable certificate. Local analyses of certainty-equivalent LQR [29] show that the suboptimality is in fact quadratic in the model error for small errors, i.e. $O(1/(N\lambda_{\text{PE}}))$. Our experiments (Fig. 3) display this faster rate. Obtaining a uniform-over-tasks version of the quadratic rate with explicit constants is left for future work.*

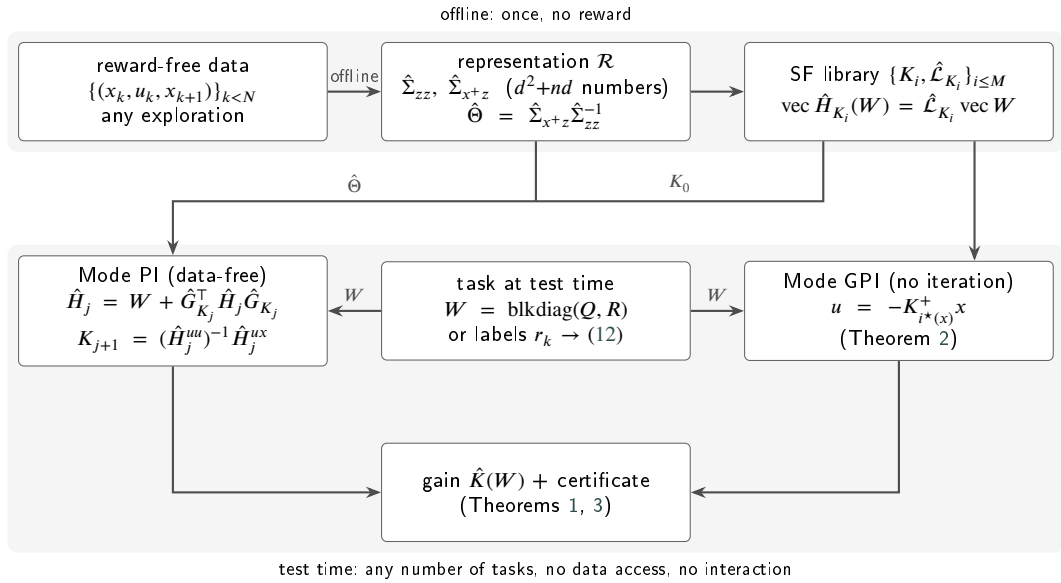


Figure 1: Zero-shot LQR. The offline phase compresses reward-free data into a fixed-size representation and, optionally, an SF library. At test time any task is mapped to a gain without touching the data or the plant.

5. Extension to continuous-time systems

Consider $\dot{x} = Ax + Bu$ with unknown (A, B) , cost $\int_0^\infty (x^\top Qx + u^\top Ru) dt$, and optimal gain $K^*(W) = R^{-1}B^\top P^*$, where P^* solves the continuous-time ARE. Kleinman's algorithm [22] iterates $(A - BK_j)^\top P_j + P_j(A - BK_j) + Q + K_j^\top RK_j = 0$, $K_{j+1} = R^{-1}B^\top P_j$. Following [18], apply an exploratory input $u = -K_b x + e(t)$ on $[t_0, t_l]$ and, for a partition $t_0 < t_1 < \dots < t_l$, form the rows ($\bar{x}$ denotes the vector with $x^\top P_x = \bar{x}^\top \text{svec}(P)$)

$$\delta_{xx} = [\bar{x}^\top]_{t_{i-1}}^{t_i}, \quad I_{xx} = \left[\int_{t_{i-1}}^{t_i} (x \otimes x)^\top d\tau \right]_i, \quad I_{xu} = \left[\int_{t_{i-1}}^{t_i} (x \otimes u)^\top d\tau \right]_i.$$

For a gain K_j , the Kleinman step is equivalent to the linear equation [18]

$$\underbrace{\begin{bmatrix} \delta_{xx}, & -2I_{xx}(I_n \otimes K_j^\top R) - 2I_{xu}(I_n \otimes R) \end{bmatrix}}_{=DT(K_j, R)} \begin{bmatrix} \text{svec}(P_j) \\ \text{vec}(K_{j+1}) \end{bmatrix} = \underbrace{-I_{xx} \text{vec}(Q + K_j^\top RK_j)}_{=Ds(K_j, W)}, \quad (19)$$

where $D := [\delta_{xx}, I_{xx}, I_{xu}]$ is the raw data matrix and $T(\cdot)$, $s(\cdot)$ are known task-dependent linear maps that do not depend on the data. Consequently the normal equations of (19) depend on the data only through the Gram matrix

$$\mathcal{R}_{CT} := D^\top D \in \mathbb{R}^{p \times p}, \quad p = \frac{n(n+1)}{2} + n^2 + nm, \quad (20)$$

whose size does not depend on the number of intervals l . Algorithm 3 is the resulting zero-shot method.

Theorem 4 (CT exactness). *Suppose the data are noise-free and satisfy the rank condition $\text{rank}[I_{xx}, I_{xu}] = n(n+1)/2 + nm$ of [18]. Then for every W with $Q > 0$, $R > 0$ and a single gain K_0 that makes $A - BK_0$ Hurwitz, the iterates of Algorithm 3 coincide with Kleinman's iterates for W ; hence every K_j is stabilizing and $K_j \rightarrow K^*(W)$ quadratically. The test-time cost per iteration is $O(p^3)$, independent of l .*

Unlike the DT case, the CT method needs a known stabilizing K_0 because Kleinman's iteration must be initialized with one. The same K_0 , however, serves all tasks, since stability of $A - BK_0$ does not depend on W .

Algorithm 3 Zero-shot LQR for CT systems**Require:** offline: integral data $(\delta_{xx}, I_{xx}, I_{xu})$; a gain K_0 that stabilizes (A, B)

- 1: **offline:** $\mathcal{R}_{CT} \leftarrow D^\top D$; discard the data
- 2: **test** (any W): $j \leftarrow 0$
- 3: **repeat**
- 4: $T \leftarrow T(K_j, R)$, $s \leftarrow s(K_j, W)$; solve $T^\top \mathcal{R}_{CT} T \theta = T^\top \mathcal{R}_{CT} s$; $K_{j+1} \leftarrow$ the K -block of θ ; $j \leftarrow j + 1$
- 5: **until** $\|K_j - K_{j-1}\| < \epsilon_{tol}$
- 6: **return** $\hat{K}(W) = K_j$

6. Simulation results

6.1. Setup

We use Rosenbrock's batch reactor [42, 11],

$$A_c = \begin{bmatrix} 1.38 & -0.2077 & 6.715 & -5.676 \\ -0.5814 & -4.29 & 0 & 0.675 \\ 1.067 & 4.273 & -6.654 & 5.893 \\ 0.048 & 4.273 & 1.343 & -2.104 \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 & 0 \\ 5.679 & 0 \\ 1.136 & -3.146 \\ 1.136 & 0 \end{bmatrix},$$

which is open-loop unstable (eigenvalues 1.99, 0.064, -5.06 , -8.67), with $n = 4$, $m = 2$, $d = 6$. The DT model is the zero-order-hold discretization with sampling time $h = 0.1$ s ($\rho(A) = 1.22$). DT data consist of episodes of 10 steps from $x_0 \sim \mathcal{N}(0, I)$ with open-loop inputs $u_k \sim \mathcal{N}(0, \sigma_e^2 I)$; no feedback, no stabilizing gain, and no reward are used. Test tasks are drawn as $Q = U \text{diag}(10^{a_i}) U^\top$, $R = V \text{diag}(10^{b_i}) V^\top$ with random orthogonal U, V and $a_i, b_i \sim \mathcal{U}(-1, 1)$, so that the weights span two orders of magnitude and are not diagonal. Ground truth is computed with the DARE/CARE of the true model. All code and data needed to reproduce every number are in the accompanying repository (code/experiments.py). [TODO: add repository link]

6.2. E1: exactness and test-time cost (DT, noise-free)

With $N = 200$ samples (20 episodes, $\sigma_e = 1$, $\lambda_{PE} = 0.138$), Mode PI was run for 200 random tasks from the same initial gain. All iterates of all tasks were stabilizing, the largest relative gain error was 5.6×10^{-14} (machine precision), and convergence took 5.6 iterations on average (at most 7). Fig. 2 shows the quadratic convergence. Table 1 compares the per-task test-time cost with per-task off-policy Q-learning on the same data: the two methods return the same gains up to 1.4×10^{-12} , but the cost of Q-learning grows linearly with N while ours does not depend on N .

6.3. E2: sample complexity and PE level (DT, noisy)

Process noise with $\sigma_w = 0.05$ was added and the exploration level was varied over $\sigma_e \in \{0.3, 1, 3\}$, which yields PE levels $\lambda_{PE} \approx 0.09, 0.15, 0.33$. For each N from 100 to 3×10^4 we averaged five independent datasets and evaluated 100 tasks. Fig. 3(a) shows that $\|\hat{\Theta} - \Theta\|$ decays as $N^{-1/2}$ and decreases with the PE level, as predicted by Lemma 2; the high-probability bound $\bar{\epsilon}_N(\delta)$ ($\delta = 0.05$) is valid and conservative by a factor of about 6–10. Fig. 3(b) reports the worst-task and median-task relative suboptimality. All $3 \times 6 \times 5 \times 100 = 9000$ returned gains stabilized the true plant. The worst-task suboptimality decays from 6.5×10^{-2} , 1.8×10^{-2} and 5.3×10^{-3} at $N = 100$ to 2.4×10^{-4} , 4.8×10^{-5} and 4.1×10^{-6} at $N = 3 \times 10^4$ for $\sigma_e = 0.3, 1, 3$ respectively, with a slope close to -1 , i.e. the quadratic regime discussed after Theorem 3.

The certificate of Theorem 3(a) is sound but conservative: it never failed when it was issued, but with the true error $\|\hat{\Theta} - \Theta\|$ in place of $\bar{\epsilon}_N$ it covered only 3%, 9% and 28% of the tasks at $N = 3 \times 10^4$, and none when $\bar{\epsilon}_N$ was used. The reason is the factor $\|\hat{P}\|/c(W)$, which ranges from 23 for $Q = I, R = I$ to more than 10^3 for the most ill-conditioned random tasks. For $Q = I, R = I$ and $\sigma_e = 3$ the certificate with $\bar{\epsilon}_N$ would be obtained for $N \gtrsim 1.7 \times 10^5$. Tightening the constants is discussed in Section 7.

6.4. E3: zero-iteration SF-GPI

With a noise-free representation ($N = 1000$), libraries of $M \in \{1, 2, 4, 8, 16\}$ gains optimal for random anchor tasks were built, and 40 new test tasks were solved in Mode GPI; the closed-loop cost of the switched policy (15) was obtained by simulation from 10 initial states. Fig. 4 and Table 2 compare SF-GPI with the best single library policy

Table 1

E1: per-task computation time (mean over 20 tasks) and the largest gap between the gains of the two methods.

N	offline (ms)	zero-shot (ms)	off-policy QL (ms)	$\max \ \hat{K} - K_{QL}\ $
200	0.05	0.37	14.5	3.1×10^{-13}
1000	0.08	0.39	73.3	1.4×10^{-12}
5000	0.14	0.39	358.5	2.2×10^{-13}

Table 2

E3: relative suboptimality (median / worst over 40 tasks).

M	best library policy	SF-GPI	GPI-init + 1 step
1	0.220 / 1.460	0.0071 / 0.377	0.0071 / 0.377
2	0.131 / 1.218	0.0047 / 0.258	0.0055 / 0.261
4	0.079 / 0.519	0.0019 / 0.056	0.0021 / 0.077
8	0.063 / 0.311	0.0014 / 0.037	0.0014 / 0.080
16	0.041 / 0.142	0.0007 / 0.013	0.0011 / 0.030

Table 3

E5: task inference from noisy reward labels (50 tasks).

label noise	median $\frac{\ \hat{W} - W\ _F}{\ W\ _F}$	median subopt.	worst subopt.
0	1.7×10^{-14}	≈ 0	3.3×10^{-14}
1%	0.020	4.6×10^{-5}	3.0×10^{-3}
10%	0.204	5.5×10^{-3}	0.168
100%	2.12	0.191	1.76

and with one Newton step from the library initialization. GPI improves on the best library policy by a factor of 30–60 in median suboptimality and reaches 0.07% with $M = 16$, at a test-time cost of 0.03–0.12 ms and without solving any equation. Every GPI closed loop was stable, as guaranteed by Theorem 2.

6.5. E4: continuous time

The CT reactor was excited with $u = -K_b x + e(t)$, where K_b is one stabilizing gain and e a sum of ten sinusoids per channel, over 2 s split into $l = 100$ intervals; the rank condition holds ($18 = 18$). The Gram representation (20) is 34×34 . For 50 random tasks started from the same $K_0 = K_b$, Algorithm 3 converged in 6.8 iterations on average, all iterates were Hurwitz-stabilizing, and the gains agree with per-task off-policy ADP [18] to 8×10^{-11} . The remaining 2×10^{-4} relative error with respect to the CARE solution comes from the numerical quadrature of the integrals, not from the method (Fig. 5(a)). Fig. 5(b) shows closed-loop responses for three very different tasks computed from the same data.

6.6. E5: tasks specified by reward labels

Instead of (Q, R) , the learner received labels $r_k = z_k^\top W z_k + v_k$ on the $N = 1000$ stored samples, with v_k Gaussian of standard deviation equal to a fraction of $\text{std}(r)$. Table 3 shows that noise-free labels identify W exactly, while 1% label noise causes a 2% weight error and a median suboptimality of 5×10^{-5} . The LQR solution is insensitive to moderate errors in the weights, so reward-label task specification is practical.

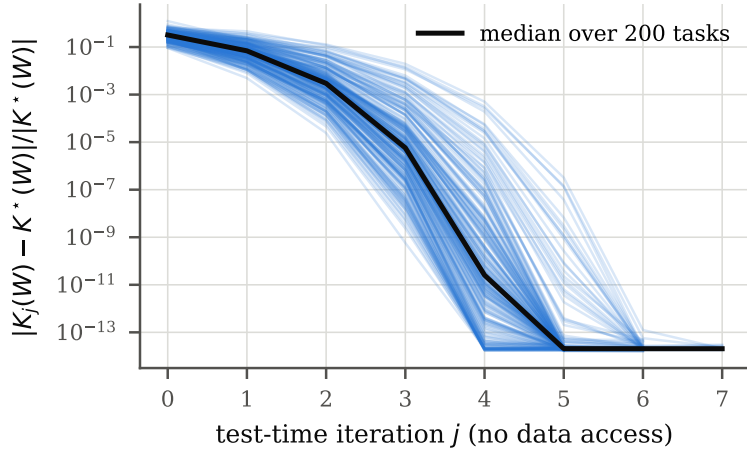


Figure 2: E1: relative gain error of the data-free test-time policy iteration for 200 random tasks from one reward-free dataset ($N = 200$). Thin lines: individual tasks; thick line: median.

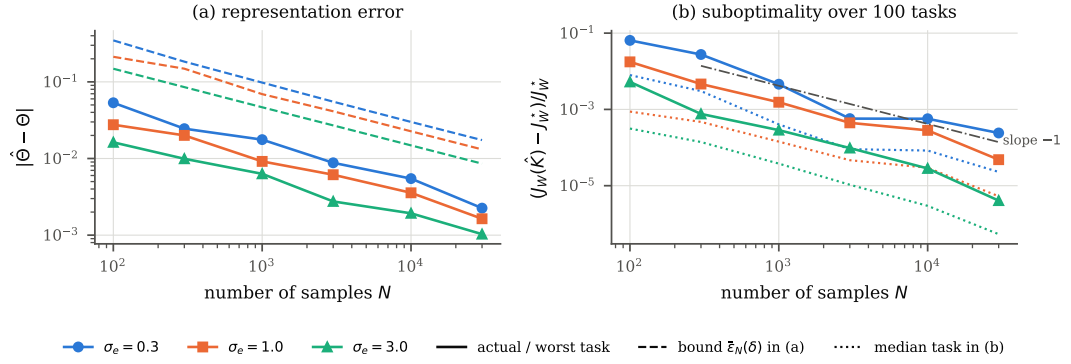


Figure 3: E2: (a) representation error versus N for three excitation levels (solid: observed, dashed: bound $\bar{\epsilon}_N(\delta)$ of Lemma 2); (b) worst-task (solid) and median-task (dotted) relative suboptimality over 100 tasks.

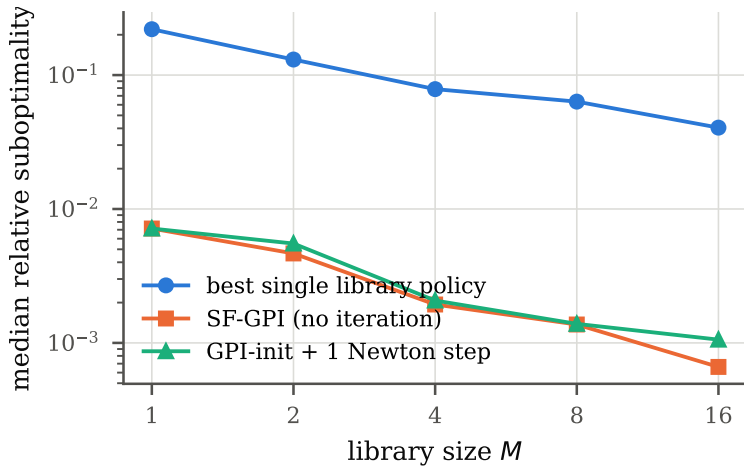


Figure 4: E3: median relative suboptimality over 40 unseen tasks versus library size.

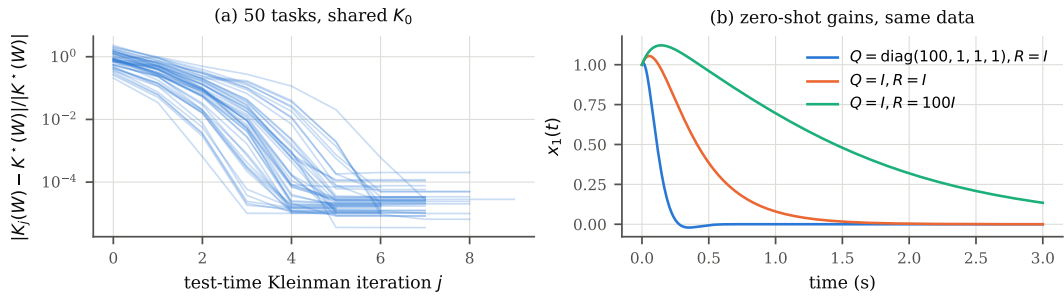


Figure 5: E4 (CT): (a) data-free Kleinman iteration for 50 tasks from one stabilizing gain; (b) first state of the closed loop under zero-shot gains for three tasks.

7. Discussion and limitations

Relation to certainty equivalence. In the linear–quadratic case the minimal zero-shot representation is equivalent to the least-squares estimate of Θ , and Mode PI returns the certainty-equivalent gain. This is not a limitation of the construction but a property of the problem. What the zero-shot viewpoint adds is (i) the task-linear successor operator, which makes policy evaluation for a new task a matrix–vector product and enables zero-iteration GPI; (ii) guarantees that hold for all tasks simultaneously with a per-task data-computable certificate; and (iii) a template for the nonlinear case, where no closed-form model is available.

Conservatism. The constants in Theorem 3 are explicit but loose, mainly through the factor $\|\hat{P}\|/c(W)$ and the linear dependence on $\bar{\epsilon}_N$. Sharper certificates could use the quadratic perturbation theory of [29], self-normalized bounds with ℓ_2 instead of per-row union bounds [33], or robust synthesis over the confidence set in the spirit of [12].

Assumptions. We assume full-state measurement, $Q > 0$, and a known noise level σ_w for the certificate. The extension to detectable $(A, Q^{1/2})$, to output feedback [32], and to H_∞ or tracking tasks, where the task additionally contains a reference or an attenuation level, are natural next steps.

Nonlinear systems. For $\dot{x} = f(x, u)$ the same two-phase structure applies with a lifted feature map ϕ (e.g., Koopman observables [23] or neural features) and costs linear in ϕ . The successor operator then becomes approximate, and the analysis requires uniformly ultimately bounded (UUB) arguments of the ADP type. This is the most important open direction.

8. Conclusion

We formulated zero-shot reinforcement learning for linear quadratic regulation and showed that successor features and the forward–backward factorization become exact and finite-dimensional in this setting. The resulting algorithm learns once from reward-free, off-policy data and returns, for any quadratic performance index revealed at test time, a stabilizing gain whose suboptimality is bounded uniformly over tasks in terms of the amount of data and the persistency-of-excitation level, or a zero-iteration GPI controller with a Lyapunov stability guarantee. Simulations on the unstable batch reactor confirm exactness with noise-free data, test-time cost independent of data length, the predicted sample complexity, and the effectiveness of SF-GPI.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

[TODO: funding information]

A. Proofs

Proof of Lemma 1

For any matrices $X \in \mathbb{R}^{d \times n}$, $Y \in \mathbb{R}^{n \times d}$, XY and YX have the same nonzero eigenvalues. With $X = \Pi_K$, $Y = \Theta$ we get $G_K = \Pi_K \Theta$ and $\Theta \Pi_K = A - BK$. From (3), $P_K = \Pi_K^\top W \Pi_K + \Pi_K^\top \Theta^\top P_K \Theta \Pi_K = \Pi_K^\top (W + \Theta^\top P_K \Theta) \Pi_K = \Pi_K^\top H_K \Pi_K$. Hence $G_K^\top H_K G_K = \Theta^\top \Pi_K^\top H_K \Pi_K \Theta = \Theta^\top P_K \Theta = H_K - W$, which is (10). Uniqueness follows from $\rho(G_K) < 1$. $\square$

Proof of Proposition 1

Along the noise-free closed loop started at $z_0 = z$ with $u_t = -Kx_t$ for $t \geq 1$, $z_t = G_K^t z$. Then $\phi(z_t) = (G_K^t z) \otimes (G_K^t z) = (G_K \otimes G_K)^t (z \otimes z)$. Since $\rho(G_K \otimes G_K) = \rho(G_K)^2 < 1$ by Lemma 1, the series converges to $(I - G_K \otimes G_K)^{-1} (z \otimes z)$, which proves (a). Then $Q_K^W(z) = \sum_t z_t^\top W z_t = \sum_t \text{vec}(W)^\top \phi(z_t) = \psi^K(z)^\top \text{vec}(W)$. Vectorizing (10) gives $\text{vec}(H_K) = \text{vec}(W) + (G_K^\top \otimes G_K^\top) \text{vec}(H_K)$, hence (b). Statement (c) is immediate from $G_K = \Pi_K \Theta$. $\square$

Proof of Theorem 1

With $w_k \equiv 0$, $X^+ = \Theta Z$, so $\hat{\Theta} = \Theta Z Z^\top (Z Z^\top)^{-1} = \Theta$. By Lemma 1, (14) evaluates $H_{K_j}(W)$ exactly, and $K_{j+1} = (H^{uu})^{-1} H^{ux} = (R + B^\top P_{K_j} B)^{-1} B^\top P_{K_j} A$ is Hewer's update. Statements (a)–(c) are Hewer's theorem [16] (see also [26]), which applies because $Q > 0$ makes $(A, Q^{1/2})$ observable. For the equivalence with off-policy Q-learning: the Q-learning evaluation of K solves in the least-squares sense $z_k^\top H z_k - (\Pi_K x_{k+1})^\top H (\Pi_K x_{k+1}) = z_k^\top W z_k$, $k < N$. Since $x_{k+1} = \Theta z_k$, this reads $z_k^\top (H - G_K^\top H G_K - W) z_k = 0$, which $H_K(W)$ satisfies, so the residual is zero. If H' is another solution, $X := H' - H_K$ is symmetric with $z_k^\top (X - G_K^\top X G_K) z_k = 0$ for all k . When the quadratic features span the symmetric matrices this forces $X - G_K^\top X G_K = 0$, hence $X = 0$ because G_K is Schur. Both methods therefore produce the same H and the same improved gain at every step. The same spanning condition makes the least-squares problem (12) uniquely solvable, with solution w . $\square$

Proof of Theorem 2

Let $Q_i := Q_{K_i}^W$ and $\ell(x, u) = x^\top Q x + u^\top R u$. Minimizing the strictly convex quadratic $Q_i(x, \cdot)$ gives $\min_u Q_i(x, u) = x^\top S_i x$ with minimizer $-K_i^+ x$, so $V(x) = \min_i \min_u Q_i(x, u)$ and $\pi_W^{\text{GPI}}(x)$ attains this minimum. Fix x , let $i = i^*(x)$, $u = \pi_W^{\text{GPI}}(x)$ and $x^+ = Ax + Bu$. By the Bellman equation of K_i ,

$$V(x) = Q_i(x, u) = \ell(x, u) + x^{+\top} P_{K_i} x^+ = \ell(x, u) + Q_i(x^+, -K_i x^+) \geq \ell(x, u) + x^{+\top} S_i x^+ \geq \ell(x, u) + V(x^+).$$

Hence $V(x^+) \leq V(x) - \ell(x, u) \leq V(x) - x^\top Q x$. For the bounds, $x^\top S_i x = \min_u \{ \ell(x, u) + (Ax + Bu)^\top P_{K_i} (Ax + Bu) \} \geq x^\top Q x$, so $V(x) \geq \lambda_{\min}(Q) \|x\|^2$, and $V(x) \leq \max_i \|S_i\| \|x\|^2$. Therefore $V(x^+) \leq (1 - \lambda_{\min}(Q) / \max_i \|S_i\|) V(x)$, which gives (a). Summing $\ell(x_t, u_t) \leq V(x_t) - V(x_{t+1})$ over $t = 0, \dots, T-1$ gives $\sum_{t < T} \ell \leq V(x_0) - V(x_T) \leq V(x_0)$; letting $T \rightarrow \infty$ and using $x^\top S_i x \leq Q_i(x, -K_i x) = x^\top P_{K_i} x$ gives (b). $\square$

Proof of Lemma 2

Write $W_N = [w_0, \dots, w_{N-1}]$, so that $X^+ = \Theta Z + W_N$ and $\hat{\Theta} - \Theta = W_N Z^\top (Z Z^\top)^{-1}$. Let $\bar{V} := Z Z^\top$ and, for each state coordinate $i \leq n$, $S_i := \sum_k z_k w_{k,i}$; the i -th row of $\hat{\Theta} - \Theta$ is $S_i^\top \bar{V}^{-1}$. Since z_k is $\mathcal{F}_k$ -measurable and $w_{k,i}$ is conditionally σ_w -sub-Gaussian, the self-normalized bound of [1, Theorem 1] with the deterministic regularizer $V = N \lambda_0 I$ gives, with probability at least $1 - \delta/n$,

$$\|S_i\|_{(\bar{V}+V)^{-1}}^2 \leq 2\sigma_w^2 \log \left(\frac{\det(\bar{V} + V)^{1/2} \det(V)^{-1/2}}{\delta/n} \right).$$

On $\{\lambda_0 I \leq \hat{\Sigma}_{zz} \leq \bar{z}^2 I\}$ we have $\bar{V} \geq V$, hence $\bar{V}^{-1} \leq 2(\bar{V} + V)^{-1}$, and $\det(\bar{V} + V) / \det(V) \leq (1 + \bar{z}^2 / \lambda_0)^d$. Therefore $\|S_i^\top \bar{V}^{-1}\|^2 \leq \|\bar{V}^{-1}\| \|S_i\|_{\bar{V}^{-1}}^2 \leq \frac{4\sigma_w^2}{N\lambda_0} \left(\frac{d}{2} \log(1 + \bar{z}^2 / \lambda_0) + \log \frac{n}{\delta} \right)$. A union bound over i and $\|\cdot\| \leq \|\cdot\|_F$ complete the proof. $\square$

Proof of Lemma 3

Let $\Pi = \Pi_K$, $M := \Pi^\top W \Pi = Q + K^\top R K$, $A_1 = \Theta_1 \Pi$, $A_2 = \Theta_2 \Pi = A_1 + E$ with $E = (\Theta_2 - \Theta_1) \Pi$. Since $W \geq cI$, $M \geq c \Pi^\top \Pi \geq cI$; since $P_1 = A_1^\top P_1 A_1 + M \geq M$, also $p_1 \geq c$, so $\varepsilon \leq 1/6$. Moreover $E^\top P_1 E \leq p_1 \varepsilon^2 \Pi^\top \Pi \leq (p_1 \varepsilon^2 / c) M$ and $P_1 \leq p_1 I \leq (p_1 / c) M$. For $\varepsilon > 0$ (the case $\varepsilon = 0$ is trivial), Young's inequality $(A_1 + E)^\top P_1 (A_1 + E) \leq (1 + \eta) A_1^\top P_1 A_1 + (1 + \eta^{-1}) E^\top P_1 E$ with $\eta = \varepsilon$ gives

$$A_2^\top P_1 A_2 \leq (1 + \varepsilon)(P_1 - M) + (1 + \varepsilon^{-1}) \frac{p_1 \varepsilon^2}{c} M \leq P_1 - \gamma M, \quad \gamma := 1 + \varepsilon - (2\varepsilon + \varepsilon^2) \frac{p_1}{c},$$

where $\varepsilon P_1 \leq (\varepsilon p_1 / c) M$ was used. With $a := \frac{13}{6} \varepsilon p_1 / c \leq \frac{13}{36}$ we get $\gamma \geq 1 - a > 0$. Hence $A_2^\top P_1 A_2 - P_1 \leq -\gamma c I < 0$, so A_2 is Schur and K stabilizes Θ_2 . Let $Y = P_1 / \gamma$. Then $A_2^\top Y A_2 + M \leq Y$, and $Y - P_2 = A_2^\top (Y - P_2) A_2 + (Y - A_2^\top Y A_2 - M)$ with the last term ≥ 0 , so $Y - P_2 = \sum_i A_2^{i\top} (Y - A_2^\top Y A_2 - M) A_2^i \geq 0$. Finally $1/\gamma \leq 1/(1 - a) \leq 1 + \frac{36}{23} a \leq 1 + 6\varepsilon p_1 / c$. $\square$

Proof of Theorem 3

By Lemma 2, with probability at least $1 - \delta$, either the PE event fails or $\|\hat{\Theta} - \Theta\| \leq \bar{\epsilon}_N =: \epsilon$; we work on the latter. This single event does not depend on W , which yields simultaneity. Fix W with $c = c(W)$, $\hat{p} = \|\hat{P}\|$, $p^* = \|P^*\|$.

(a) Since $\epsilon \leq c/(12\hat{p}) \leq c/(6\hat{p})$, Lemma 3 with $\Theta_1 = \hat{\Theta}$, $\Theta_2 = \Theta$, $K = \hat{K}$ gives $\hat{K} \in \mathcal{K}(\Theta)$ and $P_{\hat{K}} \leq (1+\alpha)\hat{P}$ with $\alpha = 6\epsilon\hat{p}/c \leq \frac{1}{2}$. Optimality gives $P^* \leq P_{\hat{K}}$, hence $p^* \leq \frac{3}{2}\hat{p}$ and $\epsilon \leq c/(12\hat{p}) \leq c/(8p^*) \leq c/(6p^*)$. Lemma 3 with $\Theta_1 = \Theta$, $\Theta_2 = \hat{\Theta}$, $K = K^*$ gives that K^* stabilizes $\hat{\Theta}$ with value $\leq (1+\beta)P^*$, $\beta = 6\epsilon p^*/c \leq 9\epsilon\hat{p}/c \leq \frac{3}{4}$. By optimality of $\hat{P}$ on $\hat{\Theta}$, $\hat{P} \leq (1+\beta)P^*$. Combining, $P^* \leq P_{\hat{K}} \leq (1+\alpha)(1+\beta)P^*$ and $\alpha+\beta+\alpha\beta \leq (6+9+4.5)\epsilon\hat{p}/c \leq 20\epsilon\hat{p}/c$, which is (17). Next, $\hat{P} - P^* \leq \beta P^* \leq \beta p^* I$ and $P^* - \hat{P} \leq \alpha \hat{P} \leq \alpha \hat{p} I$, so $\|\hat{P} - P^*\| \leq \max\{9\epsilon\hat{p} \cdot \frac{3}{2}\hat{p}, 6\epsilon\hat{p}^2\}/c \leq 14\epsilon\hat{p}^2/c$. For the gain, the cost-difference identity (obtained by telescoping $x_t^\top P^* x_t$ along the closed loop of $\hat{K}$ and using $Q^*(x, u) - x^\top P^* x = (u + K^* x)^\top (R + B^\top P^* B)(u + K^* x)$) gives

$$x^\top (P_{\hat{K}} - P^*) x = \sum_{t \geq 0} x_t^\top (\hat{K} - K^*)^\top (R + B^\top P^* B) (\hat{K} - K^*) x_t \geq \lambda_{\min}(R) \|(\hat{K} - K^*) x\|^2,$$

hence $\|\hat{K} - K^*\|^2 \leq \|P_{\hat{K}} - P^*\|/c \leq 20\epsilon\hat{p}p^*/c^2 \leq 30\epsilon\hat{p}^2/c^2$.

(b) Let $\epsilon \leq c_{\mathcal{W}}/(18\bar{p}_{\mathcal{W}})$ and $W \in \mathcal{W}$. Lemma 3 with $K = K^*$ applied from Θ to $\hat{\Theta}$ shows that K^* stabilizes $\hat{\Theta}$, so the pair $(\hat{A}, \hat{B})$ with $\hat{\Theta} = [\hat{A} \ \hat{B}]$ is stabilizable, the DARE of $\hat{\Theta}$ has a stabilizing solution $\hat{P}$ (to which Mode PI converges by Hewer's theorem), and $\hat{P} \leq (1 + 6\epsilon p^*/c)P^* \leq \frac{4}{3}P^*$. Then $\hat{p} \leq \frac{4}{3}\bar{p}_{\mathcal{W}}$, so $\epsilon \leq c/(18\bar{p}_{\mathcal{W}}) \leq c/(12\hat{p})$ and the certificate holds; by (a) the relative suboptimality is at most $20\epsilon\hat{p}/c \leq \frac{80}{3}\epsilon\bar{p}_{\mathcal{W}}/c_{\mathcal{W}} \leq 27\epsilon\bar{p}_{\mathcal{W}}/c_{\mathcal{W}}$, because $\text{tr } P_{\hat{K}} \leq (1 + \cdot) \text{tr } P^*$ follows from the matrix inequality. $\square$

Proof of Theorem 4

For each j , the exact Kleinman pair $\theta_j^* = [\text{svec}(P_j)^\top, \text{vec}(K_{j+1})^\top]^\top$ satisfies (19) with zero residual [18], i.e. $DT_j \theta_j^* = Ds_j$, so it solves the normal equations $T_j^\top D^\top DT_j \theta = T_j^\top D^\top Ds_j$. Under the rank condition, DT_j has full column rank for every K_j [18], so the normal equations have the unique solution θ_j^* . By induction the iterates of Algorithm 3 coincide with Kleinman's, and the claims follow from Kleinman's theorem [22]. Forming $T_j^\top \mathcal{R}_{CT} T_j$ and solving costs $O(p^3)$. $\square$

References

- [1] Abbasi-Yadkori, Y., Pál, D., Szepesvári, C., 2011. Improved algorithms for linear stochastic bandits, in: Advances in Neural Information Processing Systems.
- [2] Abbasi-Yadkori, Y., Szepesvári, C., 2011. Regret bounds for the adaptive control of linear quadratic systems, in: Conference on Learning Theory, pp. 1–26.
- [3] Barreto, A., Dabney, W., Munos, R., Hunt, J.J., Schaul, T., van Hasselt, H.P., Silver, D., 2017. Successor features for transfer in reinforcement learning, in: Advances in Neural Information Processing Systems.
- [4] Blier, L., Tallec, C., Ollivier, Y., 2021. Learning successor states and goal-dependent values: A mathematical viewpoint. arXiv preprint arXiv:2101.07123.
- [5] Borsa, D., Barreto, A., Quan, J., Mankowitz, D., Munos, R., van Hasselt, H., Silver, D., Schaul, T., 2019. Universal successor features approximators, in: International Conference on Learning Representations.
- [6] Bradtke, S.J., Barto, A.G., 1996. Linear least-squares algorithms for temporal difference learning. Machine Learning 22, 33–57.
- [7] Bradtke, S.J., Ydstie, B.E., Barto, A.G., 1994. Adaptive linear quadratic control using policy iteration, in: Proceedings of the American Control Conference, pp. 3475–3479.
- [8] Celi, F., Baggio, G., Pasqualetti, F., 2023. Closed-form and robust expressions for data-driven LQ control. Annual Reviews in Control 56, 100916.
- [9] Cho, J.H., Zhang, H., Du, S., Dong, R., Wu, C., 2026. Formalizing task-space complexity for zero-shot generalization. arXiv preprint arXiv:2606.20967.
- [10] Dayan, P., 1993. Improving generalization for temporal difference learning: The successor representation. Neural Computation 5, 613–624.
- [11] De Persis, C., Tesi, P., 2020. Formulas for data-driven control: Stabilization, optimality, and robustness. IEEE Transactions on Automatic Control 65, 909–924.
- [12] Dean, S., Mania, H., Matni, N., Recht, B., Tu, S., 2020. On the sample complexity of the linear quadratic regulator. Foundations of Computational Mathematics 20, 633–679.
- [13] Dörfler, F., Tesi, P., De Persis, C., 2023. On the certainty-equivalence approach to direct data-driven LQR design. IEEE Transactions on Automatic Control 68, 7989–7996.
- [14] Fazel, M., Ge, R., Kakade, S., Mesbahi, M., 2018. Global convergence of policy gradient methods for the linear quadratic regulator, in: International Conference on Machine Learning, pp. 1467–1476.

- [15] Finn, C., Abbeel, P., Levine, S., 2017. Model-agnostic meta-learning for fast adaptation of deep networks, in: International Conference on Machine Learning, pp. 1126–1135.
- [16] Hewer, G., 1971. An iterative technique for the computation of the steady state gains for the discrete optimal regulator. IEEE Transactions on Automatic Control 16, 382–384.
- [17] Jeon, S., Bewley, T., Cullen, J.M., 2024. Zero-shot reinforcement learning from low quality data, in: Advances in Neural Information Processing Systems.
- [18] Jiang, Y., Jiang, Z.P., 2012. Computational adaptive optimal control for continuous-time linear systems with completely unknown dynamics. Automatica 48, 2699–2704.
- [19] Jiang, Y., Jiang, Z.P., 2017. Robust Adaptive Dynamic Programming. Wiley-IEEE Press.
- [20] Kiumarsi, B., Lewis, F.L., Modares, H., Karimpour, A., Naghibi-Sistani, M.B., 2014. Reinforcement Q-learning for optimal tracking control of linear discrete-time systems with unknown dynamics. Automatica 50, 1167–1175.
- [21] Kiumarsi, B., Vamvoudakis, K.G., Modares, H., Lewis, F.L., 2018. Optimal and autonomous control using reinforcement learning: A survey. IEEE Transactions on Neural Networks and Learning Systems 29, 2042–2062.
- [22] Kleinman, D., 1968. On an iterative technique for Riccati equation computations. IEEE Transactions on Automatic Control 13, 114–115.
- [23] Korda, M., Mezić, I., 2018. Linear predictors for nonlinear dynamical systems: Koopman operator meets model predictive control. Automatica 93, 149–160.
- [24] Krauth, K., Tu, S., Recht, B., 2019. Finite-time analysis of approximate policy iteration for the linear quadratic regulator, in: Advances in Neural Information Processing Systems.
- [25] Lagoudakis, M.G., Parr, R., 2003. Least-squares policy iteration. Journal of Machine Learning Research 4, 1107–1149.
- [26] Lancaster, P., Rodman, L., 1995. Algebraic Riccati Equations. Oxford University Press.
- [27] Lewis, F.L., Vrabie, D., 2009. Reinforcement learning and adaptive dynamic programming for feedback control. IEEE Circuits and Systems Magazine 9, 32–50.
- [28] Lewis, F.L., Vrabie, D., Syrmos, V.L., 2012. Optimal Control. John Wiley & Sons.
- [29] Mania, H., Tu, S., Recht, B., 2019. Certainty equivalence is efficient for linear quadratic control, in: Advances in Neural Information Processing Systems.
- [30] Park, S., Kreiman, T., Levine, S., 2024. Foundation policies with Hilbert representations, in: International Conference on Machine Learning.
- [31] Possieri, C., Sassano, M., 2022. Q-learning for continuous-time linear systems: A data-driven implementation of the Kleinman algorithm. IEEE Transactions on Systems, Man, and Cybernetics: Systems 52, 6487–6497.
- [32] Rizvi, S.A.A., Lin, Z., 2019. Output feedback Q-learning control for the discrete-time linear quadratic regulator problem. IEEE Transactions on Neural Networks and Learning Systems 30, 1523–1536.
- [33] Simchowitz, M., Mania, H., Tu, S., Jordan, M.I., Recht, B., 2018. Learning without mixing: Towards a sharp analysis of linear system identification, in: Conference on Learning Theory, pp. 439–473.
- [34] Toso, L.F., Fallah, K., Stamouli, C., Pappas, G.J., Anderson, J., 2026. Multitask LQG control: Performance and generalization bounds. arXiv preprint arXiv:2604.16730 .
- [35] Toso, L.F., Zhan, D., Anderson, J., Wang, H., 2024. Meta-learning linear quadratic regulators: A policy gradient MAML approach for model-free LQR. arXiv preprint arXiv:2401.14534 .
- [36] Touati, A., Ollivier, Y., 2021. Learning one representation to optimize all rewards, in: Advances in Neural Information Processing Systems, pp. 13–23.
- [37] Touati, A., Rapin, J., Ollivier, Y., 2023. Does zero-shot reinforcement learning exist?, in: International Conference on Learning Representations.
- [38] Tu, S., Recht, B., 2019. The gap between model-based and model-free methods on the linear quadratic regulator: An asymptotic viewpoint, in: Conference on Learning Theory, pp. 3036–3083.
- [39] Vamvoudakis, K.G., 2017. Q-learning for continuous-time linear systems: A model-free infinite horizon optimal control approach. Systems & Control Letters 100, 14–20.
- [40] Vrabie, D., Pastravanu, O., Abu-Khalaf, M., Lewis, F.L., 2009. Adaptive optimal control for continuous-time linear systems based on policy iteration. Automatica 45, 477–484.
- [41] van Waarde, H.J., Eising, J., Trentelman, H.L., Camlibel, M.K., 2020. Data informativity: A new perspective on data-driven analysis and control. IEEE Transactions on Automatic Control 65, 4753–4768.
- [42] Walsh, G.C., Ye, H., Bushnell, L.G., 2002. Stability analysis of networked control systems. IEEE Transactions on Control Systems Technology 10, 438–446.
- [43] Willems, J.C., Rapisarda, P., Markovsky, I., De Moor, B.L., 2005. A note on persistency of excitation. Systems & Control Letters 54, 325–329.
- [44] Zhang, L., Fan, J., Xue, W., Lopez, V.G., Li, J., Chai, T., Lewis, F.L., 2023. Data-driven H_∞ optimal output feedback control for linear discrete-time systems based on off-policy Q-learning. IEEE Transactions on Neural Networks and Learning Systems 34, 3553–3567.